Railway planner - batch simulation report

Quantum Algorithm (Real Hardware) | 6 simulations | generated 11/7/2026, 10:31:49 PM

Setup

Algorithm	Quantum Algorithm (Real Hardware)
Budget	89
Nodes	10: Braila, Deva, Bistrita, Satu Mare, Arad, Craiova, Targu Jiu, Bucuresti, Oradea, Cluj-Napoca
Candidate routes	11: Craiova -> Bucuresti (cost 2, value 5); Cluj-Napoca -> Bucuresti (cost 19, value 21); Cluj-Napoca -> Craiova (cost 16, value 8); Arad -> Oradea (cost 3, value 8); Oradea -> Satu Mare (cost 3, value 7); Oradea -> Cluj-Napoca (cost 5, value 11); Arad -> Deva (cost 5, value 4); Cluj-Napoca -> Bistrita (cost 3, value 9); Targu Jiu -> Deva (cost 6, value 4); Targu Jiu -> Craiova (cost 2, value 8); Bucuresti -> Braila (cost 4, value 21)
Started	11/7/2026, 10:28:29 PM
Finished	11/7/2026, 10:31:20 PM

Overall statistics

Metric	Mean +/- std	Median	Min	Max	n
Approximation ratio (best sample / optimum)	1 +/- 0	1	1	1	3
Total value (valid plans)	106 +/- 0	106	106	106	3
Total cost (valid plans)	68 +/- 0	68	68	68	3
Valid sample rate	100.0% +/- 0.0%	100.0%	100.0%	100.0%	3
Connected sample rate	30.7% +/- 3.7%	29.8%	27.5%	34.8%	3
P(optimum) of the QAOA state	28.7% +/- 13.8%	36.6%	12.8%	36.8%	3
Planner runtime (s)	54.67 +/- 7.89	52.29	48.25	63.48	3
Request time (s)	55.72 +/- 7.87	53.31	49.33	64.52	3

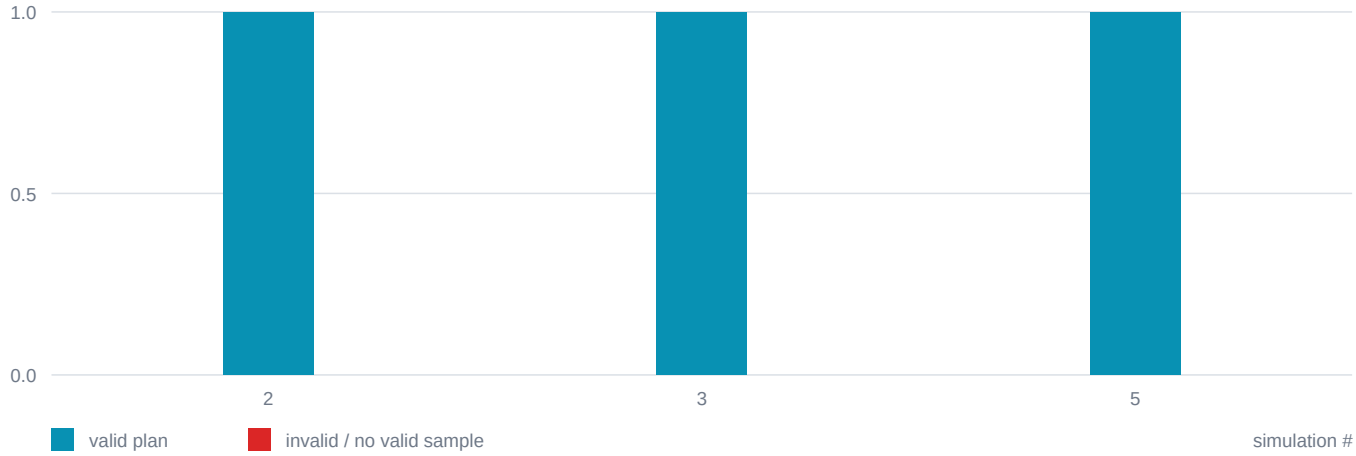
Finished 3 of 6 simulations (3 failed).

Valid plan rate: 100.0% | exact optimum found: 100.0%.

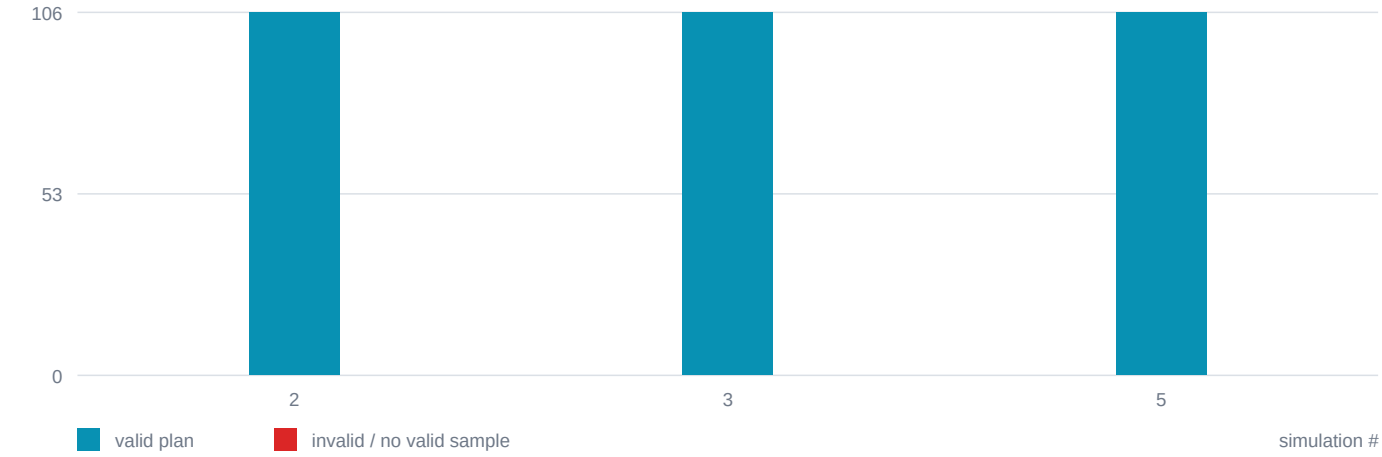
Best plan: simulation #2 (p=2 | no slack | 300 it | 1000 shots, seed 2027) - value 106, cost 68.

Charts

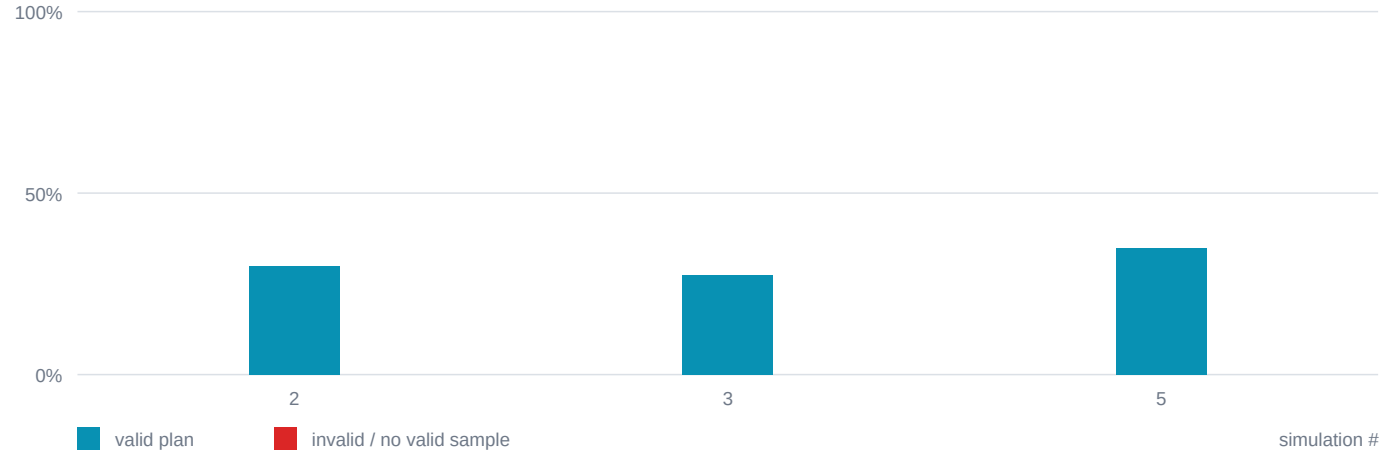
Approximation ratio per simulation



Total value per simulation



Connected sample rate per simulation



Statistics per parameter set

p	Slack	COBYLA it.	Shots	Runs	Valid	Optimum	Approx. ratio	Connected rate	Runtime (s)
2	no slack	300	1000	1	100.0%	100.0%	1	29.8%	52.29
1	no slack	500	200	1	100.0%	100.0%	1	27.5%	63.48
3	no slack	100	2048	1	100.0%	100.0%	1	34.8%	48.25

All simulations

#	p	Slack	It.	Shots	Seed	Status	Cost	Value	Optimum	Ratio	Valid rate	Conn. rate	Qubits	Time (s)
1	1	yes	300	2048	2026	error	-	-	-	-	-	-	-	1.01
2	2	no	300	1000	2027	valid	68	106	106	1	100.0%	29.8%	11	52.29
3	1	no	500	200	2028	valid	68	106	106	1	100.0%	27.5%	11	63.48
4	2	yes	300	2048	2029	error	-	-	-	-	-	-	-	0.99
5	3	no	100	2048	2030	valid	68	106	106	1	100.0%	34.8%	11	48.25
6	1	yes	300	5000	2031	error	-	-	-	-	-	-	-	0.92

Errors

#	Message
1	qaoa_only needs one qubit per route plus 7 slack qubits: 18 qubits for 11 routes, but max_qubits = 12. Options: slack = false (11 qubits); fewer candidate routes; a higher max_qubits (exact simulation of n qubits needs 2^n amplitudes: 22 qubits ~ 70 MB per state vector, 26 is the hard limit); mode = "hybrid".
4	qaoa_only needs one qubit per route plus 7 slack qubits: 18 qubits for 11 routes, but max_qubits = 12. Options: slack = false (11 qubits); fewer candidate routes; a higher max_qubits (exact simulation of n qubits needs 2^n amplitudes: 22 qubits ~ 70 MB per state vector, 26 is the hard limit); mode = "hybrid".

#	Message
6	qaoa_only needs one qubit per route plus 7 slack qubits: 18 qubits for 11 routes, but max_qubits = 12. Options: slack = false (11 qubits); fewer candidate routes; a higher max_qubits (exact simulation of n qubits needs 2^n amplitudes: 22 qubits ~ 70 MB per state vector, 26 is the hard limit); mode = "hybrid".

Circuits and QUBOs of the batch

#	p	Slack	Qubits	Depth	2q gates	Gate counts	QUBO lin./quad.	QUBO energy range	Expected energy
2	2	no	11	36	110	rzz 110, rz 22, rx 22, h 11, measure 11	11 / 55	-122.59 ... -7.29	-118.407
3	1	no	11	23	55	rzz 55, h 11, rz 11, rx 11, measure 11	11 / 55	-122.59 ... -7.29	-113.427
5	3	no	11	49	165	rzz 165, rz 33, rx 33, h 11, measure 11	11 / 55	-122.59 ... -7.29	-120.739

Simulation #2

p = 2 | no slack | 300 COBYLA iterations | 1000 shots | seed 2027 | valid plan

Intermediate values

#	Step	Value
1	**Planner result**	Connected network returned within the 89 budget.
2	**QAOA configuration**	ibm-qpu | 11 qubits | 11 route + 0 slack | p = 2 | unbalanced penalization (no slack) | 300 COBYLA iterations | 1000 shots
3	**Penalty weights**	unbalanced penalization (no slack) | lambda1 1 | lambda2 0.01 | mu (conflicts) 107 | beta (city cover) 0.25 | budget 89 units
4	**Optimised angles**	gamma [-1.0667, -2.719] | beta [0.8285, 0.268]
5	**Angle training (COBYLA)**	8 optimiser runs, 2484 energy evaluations | warm start auto | selected interp / INTERP
6	**Optimiser run 1**	interp | p = 1 | start grid scan | E -113.421 -> -113.427 | 384 evaluations | gamma [-0.9898] | beta [0.7787]
7	**Optimiser run 2 (selected)**	interp | p = 2 | start INTERP | E -82.223 -> -118.407 | 300 evaluations | gamma [-1.0667, -2.719] | beta [0.8285, 0.268]
8	**Optimiser run 3**	previous | p = 2 | start ramp dt=0.02 | E -76.836 -> -113.72 | 300 evaluations | gamma [-0.7367, -0.559] | beta [0.5139, 0.327]
9	**Optimiser run 4**	previous | p = 2 | start ramp dt=0.05 | E -83.125 -> -117.197 | 300 evaluations | gamma [-0.9195, -2.0581] | beta [0.8623, 0.2445]
10	**Optimiser run 5**	previous | p = 2 | start ramp dt=0.1 | E -92.976 -> -118.031 | 300 evaluations | gamma [-1.0053, -2.3985] | beta [0.8521, 0.2586]
11	**Optimiser run 6**	previous | p = 2 | start random | E -97.022 -> -113.722 | 300 evaluations | gamma [-0.6005, -0.4182] | beta [-0.273, 0.9903]
12	**Optimiser run 7**	previous | p = 2 | start random | E -67.981 -> -118.265 | 300 evaluations | gamma [-1.0384, -2.5358] | beta [0.8494, 0.2619]
13	**Optimiser run 8**	previous | p = 2 | start random | E -88.349 -> -117.95 | 300 evaluations | gamma [-0.8498, -1.4614] | beta [-0.8878, 1.291]
14	**QAOA circuit**	11 qubits | depth 36 | 110 two-qubit gates
15	**Expected energy**	-118.407
16	**P(valid) of the QAOA state**	100.00% (uniform 100.00%)
17	**P(connected) of the QAOA state**	86.48%
18	**P(optimum) of the QAOA state**	36.79%
19	**Valid sample rate**	100.00%
20	**Connected sample rate**	29.80%
21	**Best sampled value**	106
22	**Exact optimum value**	106
23	**Approximation ratio**	1 | exact optimum found
24	**Reference optimum (exhaustive check)**	value 106 | cost 68 | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
25	**Top sample 1**	67 shots | value 102 | cost 62 | valid, connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Craiova, Bucuresti -> Braila
26	**Top sample 2**	53 shots | value 106 | cost 68 | valid, connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
27	**Top sample 3**	43 shots | value 102 | cost 63 | valid, connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
28	**Top sample 4**	36 shots | value 98 | cost 57 | valid, not connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Cluj-Napoca -> Bistrita, Targu Jiu -> Craiova, Bucuresti -> Braila
29	**Top sample 5**	27 shots | value 95 | cost 59 | valid, not connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Craiova, Bucuresti -> Braila
30	**Top sample 6**	25 shots | value 101 | cost 66 | valid, connected | Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
31	**Top sample 7**	15 shots | value 97 | cost 61 | valid, connected | Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
32	**Top sample 8**	12 shots | value 95 | cost 60 | valid, not connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Cluj-Napoca, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila

#	Step	Value
33	**Planner timing**	penalty tuning 1.342 s | angle training 5.688 s | sampling 45.199 s
34	**Hardware execution**	ibm_aachen | status done | 1000 shots | job dbnojvcnqjds73evhnqg | auto layout | compiled depth 419 | 365 two-qubit gates | estimated success 51.04%
35	**Cost breakdown**	Quantum 68 | connectors 0 | extra 0
36	**quantum route**	Craiova -> Bucuresti | cost 2 | reward +5
37	**quantum route**	Cluj-Napoca -> Bucuresti | cost 19 | reward +21
38	**quantum route**	Cluj-Napoca -> Craiova | cost 16 | reward +8
39	**quantum route**	Arad -> Oradea | cost 3 | reward +8
40	**quantum route**	Oradea -> Satu Mare | cost 3 | reward +7
41	**quantum route**	Oradea -> Cluj-Napoca | cost 5 | reward +11
42	**quantum route**	Arad -> Deva | cost 5 | reward +4
43	**quantum route**	Cluj-Napoca -> Bistrita | cost 3 | reward +9
44	**quantum route**	Targu Jiu -> Deva | cost 6 | reward +4
45	**quantum route**	Targu Jiu -> Craiova | cost 2 | reward +8
46	**quantum route**	Bucuresti -> Braila | cost 4 | reward +21

QUBO

Budget encoding	unbalanced penalization (no slack)
Offset	-7.29
Energy range	-122.59 ... -7.29
Terms	11 linear, 55 quadratic
Variables	x0 = Craiova -> Bucuresti; x1 = Cluj-Napoca -> Bucuresti; x2 = Cluj-Napoca -> Craiova; x3 = Arad -> Oradea; x4 = Oradea -> Satu Mare; x5 = Oradea -> Cluj-Napoca; x6 = Arad -> Deva; x7 = Cluj-Napoca -> Bistrita; x8 = Targu Jiu -> Deva; x9 = Targu Jiu -> Craiova; x10 = Bucuresti -> Braila

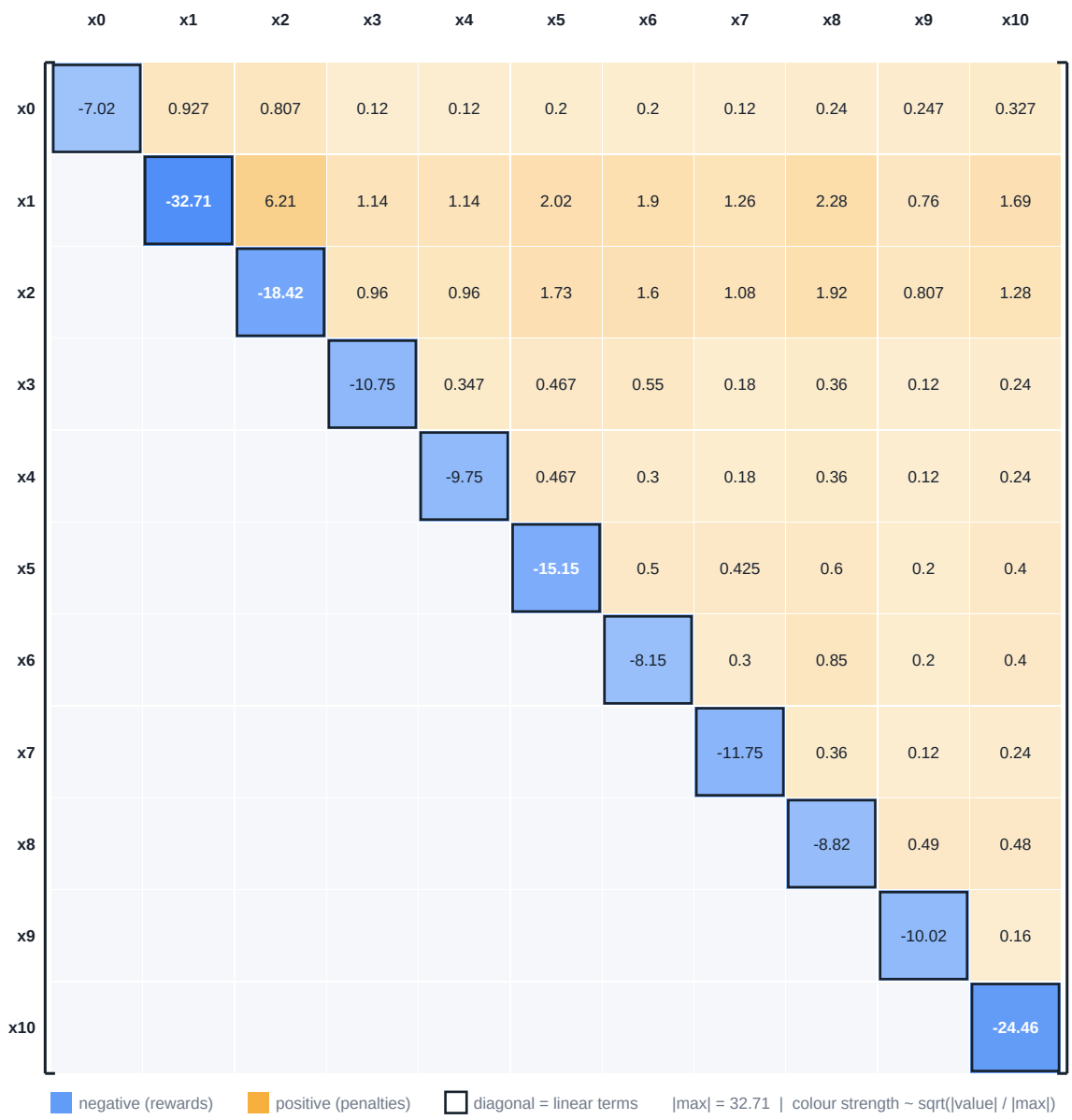
Formula

```
E = -(5 x0 + 21 x1 + 8 x2 + 8 x3 + 7 x4 + 11 x5 + 4 x6 + 9 x7 + 4 x8 + 8 x9 + 21 x10) - 1*(89 - (2 x0 + 19 x1 + 16 x2 + 3 x3 + 3 x4 + 5 x5 + 5 x6 + 3 x7 + 6 x8 + 2 x9 + 4 x10))^2 + 0.25((1 - x0 - x1 - x10 + 0.666667*(x0*x1 + x0*x10 + x1*x10)) + (1 - x1 - x2 - x5 - x7 + 0.5*(x1*x2 + x1*x5 + x1*x7 + x2*x5 + x2*x7 + x5*x7)) + (1 - x0 - x2 - x9 + 0.666667*(x0*x2 + x0*x9 + x2*x9)) + (1 - x3 - x4 - x5 + 0.666667*(x3*x4 + x3*x5 + x4*x5)) + (1 - x10) + (1 - x3 - x6 + 1*(x3*x6)) + (1 - x4) + (1 - x8 - x9 + 1*(x8*x9)) + (1 - x7) + (1 - x6 - x8 + 1*(x6*x8)))
```

Term	Coefficient	Term	Coefficient	Term	Coefficient
x0	-7.02	x1	-32.71	x2	-18.42
x3	-10.75	x4	-9.75	x5	-15.15
x6	-8.15	x7	-11.75	x8	-8.82
x9	-10.02	x10	-24.46	x0*x1	0.9267
x0*x2	0.8067	x0*x3	0.12	x0*x4	0.12
x0*x5	0.2	x0*x6	0.2	x0*x7	0.12
x0*x8	0.24	x0*x9	0.2467	x0*x10	0.3267
x1*x2	6.205	x1*x3	1.14	x1*x4	1.14
x1*x5	2.025	x1*x6	1.9	x1*x7	1.265
x1*x8	2.28	x1*x9	0.76	x1*x10	1.6867
x2*x3	0.96	x2*x4	0.96	x2*x5	1.725
x2*x6	1.6	x2*x7	1.085	x2*x8	1.92
x2*x9	0.8067	x2*x10	1.28	x3*x4	0.3467
x3*x5	0.4667	x3*x6	0.55	x3*x7	0.18
x3*x8	0.36	x3*x9	0.12	x3*x10	0.24
x4*x5	0.4667	x4*x6	0.3	x4*x7	0.18
x4*x8	0.36	x4*x9	0.12	x4*x10	0.24
x5*x6	0.5	x5*x7	0.425	x5*x8	0.6
x5*x9	0.2	x5*x10	0.4	x6*x7	0.3
x6*x8	0.85	x6*x9	0.2	x6*x10	0.4
x7*x8	0.36	x7*x9	0.12	x7*x10	0.24
x8*x9	0.49	x8*x10	0.48	x9*x10	0.16

QUBO matrix (11 x 11)

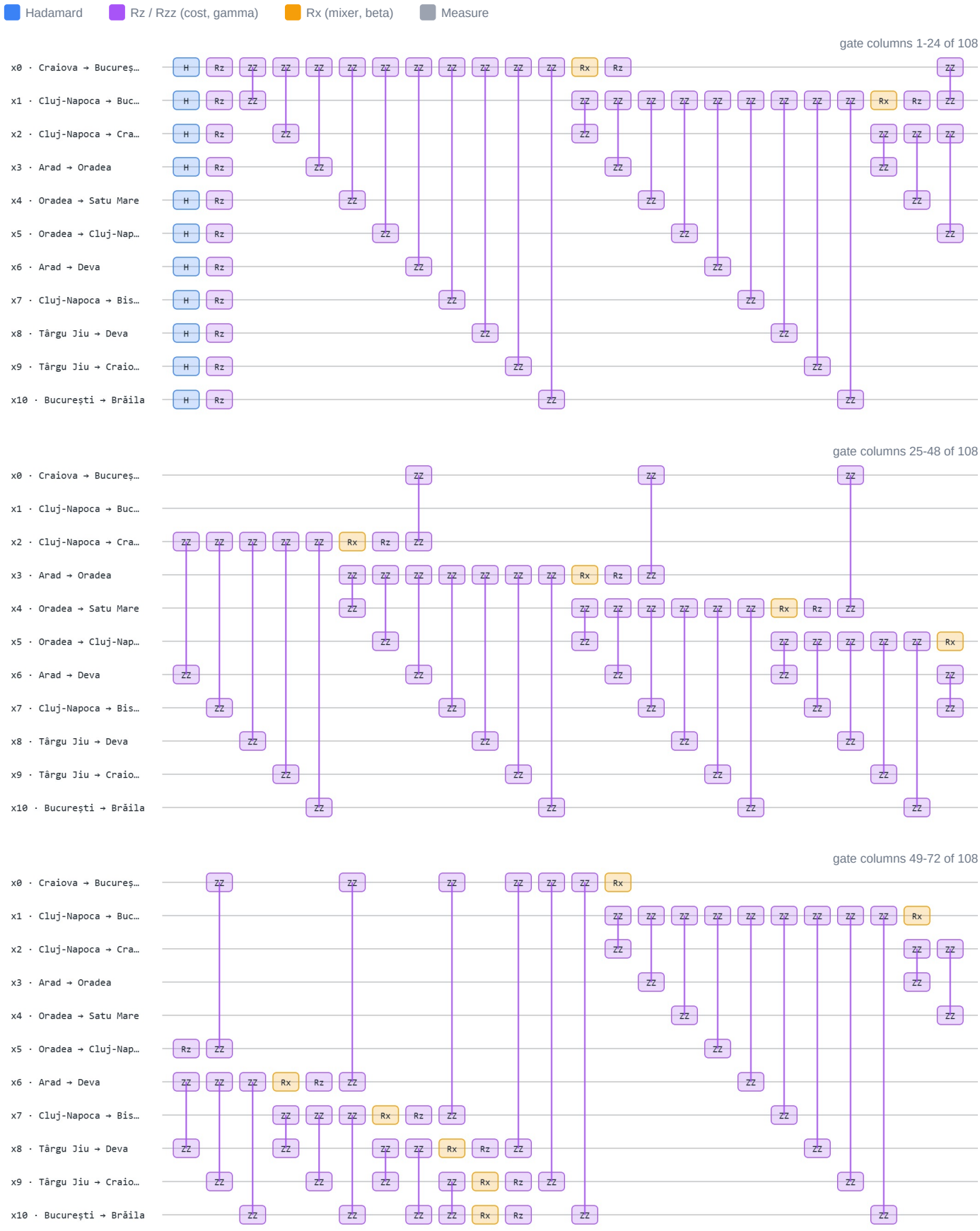
upper-triangular Q: row/column order = matrix_variables, diagonal Q[i][i] = linear coefficient of y_i, Q[i][j] (i < j) = coefficient of y_i*y_j, below the diagonal 0; E(y) = offset + y^T Q y (bits, so y_i^2 = y_i)



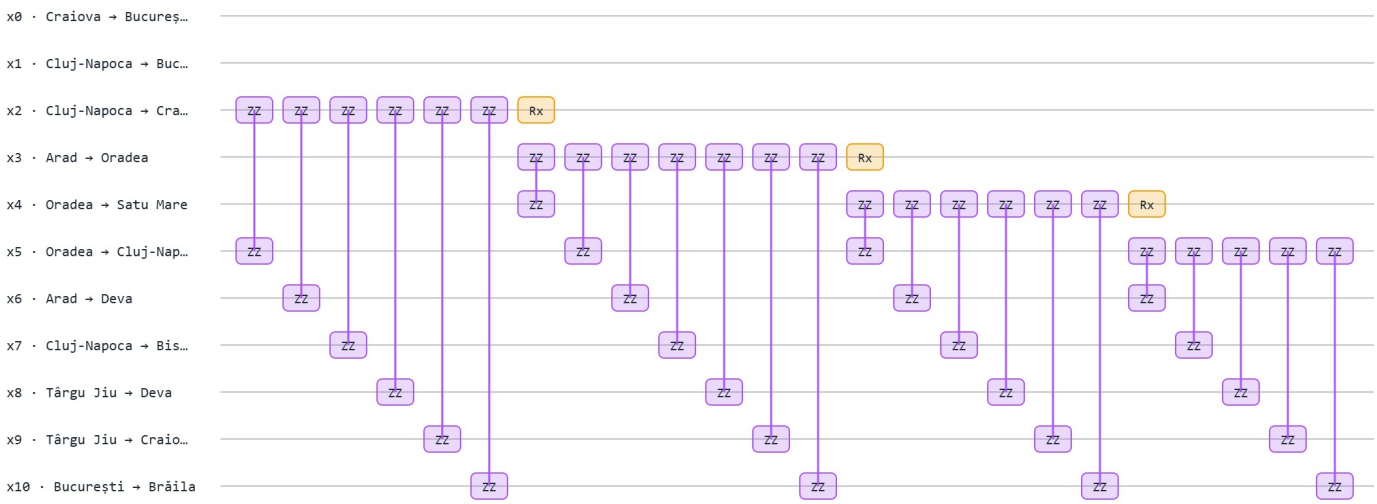
QAOA circuit

11 qubits | depth 36 | 110 two-qubit gates | rzz 110, rz 22, rx 22, h 11, measure 11

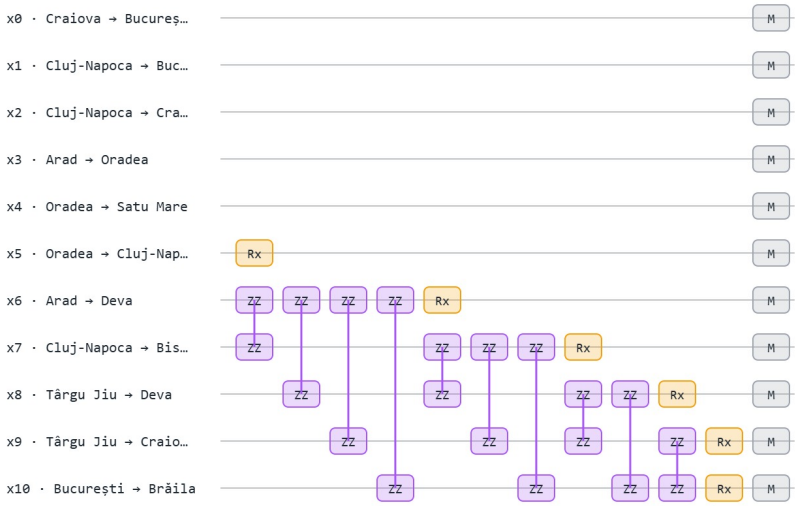
gamma = [-1.0667, -2.7190] beta = [0.8285, 0.2680]


Hadamard
Rz / Rzz (cost, gamma)
Rx (mixer, beta)
Measure
gate columns 1-24 of 108
x0 · Craiova → Bucureș…
x1 · Cluj-Napoca → Buc…
x2 · Cluj-Napoca → Cra…
x3 · Arad → Oradea
x4 · Oradea → Satu Mare
x5 · Oradea → Cluj-Nap…
x6 · Arad → Deva
x7 · Cluj-Napoca → Bis…
x8 · Târgu Jiu → Deva
x9 · Târgu Jiu → Craio…
x10 · București → Brăila
gate columns 25-48 of 108
gate columns 49-72 of 108

gate columns 73-96 of 108



gate columns 97-108 of 108



Simulation #3

p = 1 | no slack | 500 COBYLA iterations | 200 shots | seed 2028 | valid plan

Intermediate values

#	Step	Value
1	**Planner result**	Connected network returned within the 89 budget.
2	**QAOA configuration**	ibm-qpu | 11 qubits | 11 route + 0 slack | p = 1 | unbalanced penalization (no slack) | 500 COBYLA iterations | 200 shots
3	**Penalty weights**	unbalanced penalization (no slack) | lambda1 1 | lambda2 0.01 | mu (conflicts) 107 | beta (city cover) 0.25 | budget 89 units
4	**Optimised angles**	gamma [-0.9899] | beta [0.7786]
5	**Angle training (COBYLA)**	7 optimiser runs, 627 energy evaluations | warm start auto | selected previous / ramp dt=0.02
6	**Optimiser run 1**	interp | p = 1 | start grid scan | E -113.421 -> -113.427 | 384 evaluations | gamma [-0.9898] | beta [0.7787]
7	**Optimiser run 2 (selected)**	previous | p = 1 | start ramp dt=0.02 | E -76.041 -> -113.427 | 40 evaluations | gamma [-0.9899] | beta [0.7786]
8	**Optimiser run 3**	previous | p = 1 | start ramp dt=0.05 | E -79.316 -> -113.427 | 41 evaluations | gamma [-0.99] | beta [0.7787]
9	**Optimiser run 4**	previous | p = 1 | start ramp dt=0.1 | E -87.81 -> -113.427 | 37 evaluations | gamma [-0.9899] | beta [0.7785]
10	**Optimiser run 5**	previous | p = 1 | start random | E -79.801 -> -113.427 | 38 evaluations | gamma [-0.9898] | beta [0.7785]
11	**Optimiser run 6**	previous | p = 1 | start random | E -74.911 -> -113.427 | 45 evaluations | gamma [0.9899] | beta [2.3631]
12	**Optimiser run 7**	previous | p = 1 | start random | E -61.52 -> -113.427 | 42 evaluations | gamma [-0.9899] | beta [0.7787]
13	**QAOA circuit**	11 qubits | depth 23 | 55 two-qubit gates
14	**Expected energy**	-113.427
15	**P(valid) of the QAOA state**	100.00% (uniform 100.00%)
16	**P(connected) of the QAOA state**	47.61%
17	**P(optimum) of the QAOA state**	12.76%
18	**Valid sample rate**	100.00%
19	**Connected sample rate**	27.50%
20	**Best sampled value**	106
21	**Exact optimum value**	106
22	**Approximation ratio**	1 | exact optimum found
23	**Reference optimum (exhaustive check)**	value 106 | cost 68 | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
24	**Top sample 1**	9 shots | value 106 | cost 68 | valid, connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
25	**Top sample 2**	7 shots | value 102 | cost 62 | valid, connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Craiova, Bucuresti -> Braila
26	**Top sample 3**	6 shots | value 98 | cost 65 | valid, connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
27	**Top sample 4**	5 shots | value 101 | cost 66 | valid, connected | Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
28	**Top sample 5**	5 shots | value 102 | cost 63 | valid, connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
29	**Top sample 6**	5 shots | value 92 | cost 63 | valid, not connected | Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
30	**Top sample 7**	4 shots | value 85 | cost 49 | valid, connected | Craiova -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
31	**Top sample 8**	3 shots | value 78 | cost 36 | valid, not connected | Cluj-Napoca -> Bucuresti, Arad -> Oradea, Oradea -> Cluj-Napoca, Cluj-Napoca -> Bistrita, Targu Jiu -> Craiova, Bucuresti -> Braila
32	**Planner timing**	penalty tuning 1.313 s | angle training 0.574 s | sampling 61.554 s
33	**Hardware execution**	ibm_aachen | status done | 200 shots | job dbnokb2ft1mc73fl3k1g | auto layout | compiled depth 319 | 186 two-qubit gates | estimated success 69.59%
34	**Cost breakdown**	Quantum 68 | connectors 0 | extra 0
35	**quantum route**	Craiova -> Bucuresti | cost 2 | reward +5
36	**quantum route**	Cluj-Napoca -> Bucuresti | cost 19 | reward +21

#	Step	Value
37	**quantum route**	Cluj-Napoca -> Craiova | cost 16 | reward +8
38	**quantum route**	Arad -> Oradea | cost 3 | reward +8
39	**quantum route**	Oradea -> Satu Mare | cost 3 | reward +7
40	**quantum route**	Oradea -> Cluj-Napoca | cost 5 | reward +11
41	**quantum route**	Arad -> Deva | cost 5 | reward +4
42	**quantum route**	Cluj-Napoca -> Bistrita | cost 3 | reward +9
43	**quantum route**	Targu Jiu -> Deva | cost 6 | reward +4
44	**quantum route**	Targu Jiu -> Craiova | cost 2 | reward +8
45	**quantum route**	Bucuresti -> Braila | cost 4 | reward +21

QUBO

Budget encoding	unbalanced penalization (no slack)
Offset	-7.29
Energy range	-122.59 ... -7.29
Terms	11 linear, 55 quadratic
Variables	x0 = Craiova -> Bucuresti; x1 = Cluj-Napoca -> Bucuresti; x2 = Cluj-Napoca -> Craiova; x3 = Arad -> Oradea; x4 = Oradea -> Satu Mare; x5 = Oradea -> Cluj-Napoca; x6 = Arad -> Deva; x7 = Cluj-Napoca -> Bistrita; x8 = Targu Jiu -> Deva; x9 = Targu Jiu -> Craiova; x10 = Bucuresti -> Braila

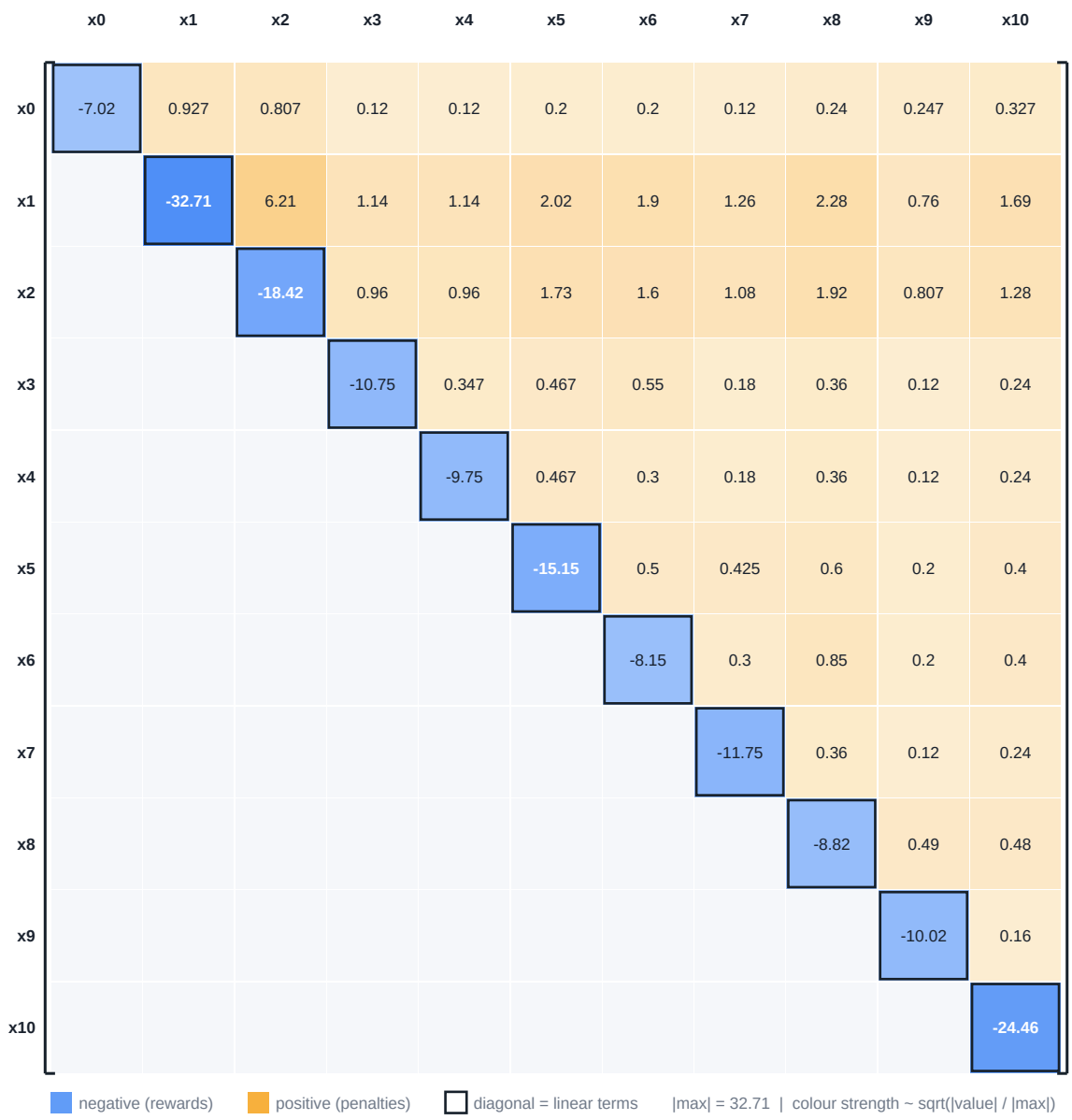
Formula

```
E = -(5 x0 + 21 x1 + 8 x2 + 8 x3 + 7 x4 + 11 x5 + 4 x6 + 9 x7 + 4 x8 + 8 x9 + 21 x10) - 1*(89 - (2 x0 + 19 x1 + 16 x2 + 3 x3 + 3 x4 + 5 x5 + 5 x6 + 3 x7 + 6 x8 + 2 x9 + 4 x10))^2 + 0.25((1 - x0 - x1 - x10 + 0.666667*(x0*x1 + x0*x10 + x1*x10)) + (1 - x1 - x2 - x5 - x7 + 0.5*(x1*x2 + x1*x5 + x1*x7 + x2*x5 + x2*x7 + x5*x7)) + (1 - x0 - x2 - x9 + 0.666667*(x0*x2 + x0*x9 + x2*x9)) + (1 - x3 - x4 - x5 + 0.666667*(x3*x4 + x3*x5 + x4*x5)) + (1 - x10) + (1 - x3 - x6 + 1*(x3*x6)) + (1 - x4) + (1 - x8 - x9 + 1*(x8*x9)) + (1 - x7) + (1 - x6 - x8 + 1*(x6*x8)))
```

Term	Coefficient	Term	Coefficient	Term	Coefficient
x0	-7.02	x1	-32.71	x2	-18.42
x3	-10.75	x4	-9.75	x5	-15.15
x6	-8.15	x7	-11.75	x8	-8.82
x9	-10.02	x10	-24.46	x0*x1	0.9267
x0*x2	0.8067	x0*x3	0.12	x0*x4	0.12
x0*x5	0.2	x0*x6	0.2	x0*x7	0.12
x0*x8	0.24	x0*x9	0.2467	x0*x10	0.3267
x1*x2	6.205	x1*x3	1.14	x1*x4	1.14
x1*x5	2.025	x1*x6	1.9	x1*x7	1.265
x1*x8	2.28	x1*x9	0.76	x1*x10	1.6867
x2*x3	0.96	x2*x4	0.96	x2*x5	1.725
x2*x6	1.6	x2*x7	1.085	x2*x8	1.92
x2*x9	0.8067	x2*x10	1.28	x3*x4	0.3467
x3*x5	0.4667	x3*x6	0.55	x3*x7	0.18
x3*x8	0.36	x3*x9	0.12	x3*x10	0.24
x4*x5	0.4667	x4*x6	0.3	x4*x7	0.18
x4*x8	0.36	x4*x9	0.12	x4*x10	0.24
x5*x6	0.5	x5*x7	0.425	x5*x8	0.6
x5*x9	0.2	x5*x10	0.4	x6*x7	0.3
x6*x8	0.85	x6*x9	0.2	x6*x10	0.4
x7*x8	0.36	x7*x9	0.12	x7*x10	0.24
x8*x9	0.49	x8*x10	0.48	x9*x10	0.16

QUBO matrix (11 x 11)

upper-triangular Q: row/column order = matrix_variables, diagonal Q[i][i] = linear coefficient of y_i, Q[i][j] (i < j) = coefficient of y_i*y_j, below the diagonal 0; E(y) = offset + y^T Q y (bits, so y_i^2 = y_i)



QAOA circuit

11 qubits | depth 23 | 55 two-qubit gates | rzz 55, h 11, rz 11, rx 11, measure 11

gamma = [-0.9899] beta = [0.7786]


Hadamard
Rz / Rzz (cost, gamma)
Rx (mixer, beta)
Measure
gate columns 1-24 of 59
x0 · Craiova → Bucureș…
x1 · Cluj-Napoca → Buc…
x2 · Cluj-Napoca → Cra…
x3 · Arad → Oradea
x4 · Oradea → Satu Mare
x5 · Oradea → Cluj-Nap…
x6 · Arad → Deva
x7 · Cluj-Napoca → Bis…
x8 · Târgu Jiu → Deva
x9 · Târgu Jiu → Craio…
x10 · București → Brăila
gate columns 25-48 of 59
gate columns 49-59 of 59

Simulation #5

p = 3 | no slack | 100 COBYLA iterations | 2048 shots | seed 2030 | valid plan

Intermediate values

#	Step	Value
1	**Planner result**	Connected network returned within the 89 budget.
2	**QAOA configuration**	ibm-qpu | 11 qubits | 11 route + 0 slack | p = 3 | unbalanced penalization (no slack) | 100 COBYLA iterations | 2048 shots
3	**Penalty weights**	unbalanced penalization (no slack) | lambda1 1 | lambda2 0.01 | mu (conflicts) 107 | beta (city cover) 0.25 | budget 89 units
4	**Optimised angles**	gamma [-0.687, -1.4704, -2.1087] | beta [0.8459, 0.3756, 0.2695]
5	**Angle training (COBYLA)**	9 optimiser runs, 1184 energy evaluations | warm start auto | selected interp / INTERP
6	**Optimiser run 1**	interp | p = 1 | start grid scan | E -113.421 -> -113.427 | 384 evaluations | gamma [-0.9898] | beta [0.7787]
7	**Optimiser run 2**	interp | p = 2 | start INTERP | E -82.223 -> -116.246 | 100 evaluations | gamma [-0.8011, -1.7132] | beta [0.8513, 0.2615]
8	**Optimiser run 3 (selected)**	interp | p = 3 | start INTERP | E -115.263 -> -120.739 | 100 evaluations | gamma [-0.687, -1.4704, -2.1087] | beta [0.8459, 0.3756, 0.2695]
9	**Optimiser run 4**	previous | p = 3 | start ramp dt=0.02 | E -77.786 -> -113.5 | 100 evaluations | gamma [-0.6656, -0.3107, -0.2023] | beta [0.2705, 0.3078, 0.2481]
10	**Optimiser run 5**	previous | p = 3 | start ramp dt=0.05 | E -86.699 -> -114.781 | 100 evaluations | gamma [-0.512, -0.5303, -0.7727] | beta [0.5706, 0.2841, 0.2259]
11	**Optimiser run 6**	previous | p = 3 | start ramp dt=0.1 | E -94.588 -> -114.215 | 100 evaluations | gamma [0.4203, -0.4735, -0.4266] | beta [0.8373, 0.6329, 0.5748]
12	**Optimiser run 7**	previous | p = 3 | start random | E -50.631 -> -114.474 | 100 evaluations | gamma [-0.498, -0.5481, -0.551] | beta [0.6316, 0.1291, 0.3082]
13	**Optimiser run 8**	previous | p = 3 | start random | E -51.552 -> -114.2 | 100 evaluations | gamma [-0.804, 0.1564, -0.4541] | beta [0.85, 1.6314, 1.3564]
14	**Optimiser run 9**	previous | p = 3 | start random | E -71.443 -> -114.152 | 100 evaluations | gamma [0.3643, -0.5219, -0.2174] | beta [1.2886, 0.5403, 0.4034]
15	**QAOA circuit**	11 qubits | depth 49 | 165 two-qubit gates
16	**Expected energy**	-120.739
17	**P(valid) of the QAOA state**	100.00% (uniform 100.00%)
18	**P(connected) of the QAOA state**	91.34%
19	**P(optimum) of the QAOA state**	36.58%
20	**Valid sample rate**	100.00%
21	**Connected sample rate**	34.77%
22	**Best sampled value**	106
23	**Exact optimum value**	106
24	**Approximation ratio**	1 | exact optimum found
25	**Reference optimum (exhaustive check)**	value 106 | cost 68 | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
26	**Top sample 1**	137 shots | value 106 | cost 68 | valid, connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
27	**Top sample 2**	100 shots | value 102 | cost 62 | valid, connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Craiova, Bucuresti -> Braila
28	**Top sample 3**	90 shots | value 98 | cost 52 | valid, connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
29	**Top sample 4**	62 shots | value 102 | cost 63 | valid, connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
30	**Top sample 5**	39 shots | value 85 | cost 49 | valid, connected | Craiova -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
31	**Top sample 6**	38 shots | value 98 | cost 57 | valid, not connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Cluj-Napoca -> Bistrita, Targu Jiu -> Craiova, Bucuresti -> Braila
32	**Top sample 7**	34 shots | value 98 | cost 65 | valid, connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca -> Craiova, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Cluj-Napoca -> Bistrita, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
33	**Top sample 8**	32 shots | value 97 | cost 65 | valid, not connected | Craiova -> Bucuresti, Cluj-Napoca -> Bucuresti, Cluj-Napoca ->

#	Step	Value
		Craiova, Arad -> Oradea, Oradea -> Satu Mare, Oradea -> Cluj-Napoca, Arad -> Deva, Targu Jiu -> Deva, Targu Jiu -> Craiova, Bucuresti -> Braila
34	**Planner timing**	penalty tuning 1.426 s | angle training 2.813 s | sampling 43.94 s
35	**Hardware execution**	ibm_aachen | status done | 2048 shots | job dbnokrsnqjds73evhok0 | auto layout | compiled depth 722 | 599 two-qubit gates | estimated success 33.78%
36	**Cost breakdown**	Quantum 68 | connectors 0 | extra 0
37	**quantum route**	Craiova -> Bucuresti | cost 2 | reward +5
38	**quantum route**	Cluj-Napoca -> Bucuresti | cost 19 | reward +21
39	**quantum route**	Cluj-Napoca -> Craiova | cost 16 | reward +8
40	**quantum route**	Arad -> Oradea | cost 3 | reward +8
41	**quantum route**	Oradea -> Satu Mare | cost 3 | reward +7
42	**quantum route**	Oradea -> Cluj-Napoca | cost 5 | reward +11
43	**quantum route**	Arad -> Deva | cost 5 | reward +4
44	**quantum route**	Cluj-Napoca -> Bistrita | cost 3 | reward +9
45	**quantum route**	Targu Jiu -> Deva | cost 6 | reward +4
46	**quantum route**	Targu Jiu -> Craiova | cost 2 | reward +8
47	**quantum route**	Bucuresti -> Braila | cost 4 | reward +21

QUBO

Budget encoding	unbalanced penalization (no slack)
Offset	-7.29
Energy range	-122.59 ... -7.29
Terms	11 linear, 55 quadratic
Variables	x0 = Craiova -> Bucuresti; x1 = Cluj-Napoca -> Bucuresti; x2 = Cluj-Napoca -> Craiova; x3 = Arad -> Oradea; x4 = Oradea -> Satu Mare; x5 = Oradea -> Cluj-Napoca; x6 = Arad -> Deva; x7 = Cluj-Napoca -> Bistrita; x8 = Targu Jiu -> Deva; x9 = Targu Jiu -> Craiova; x10 = Bucuresti -> Braila

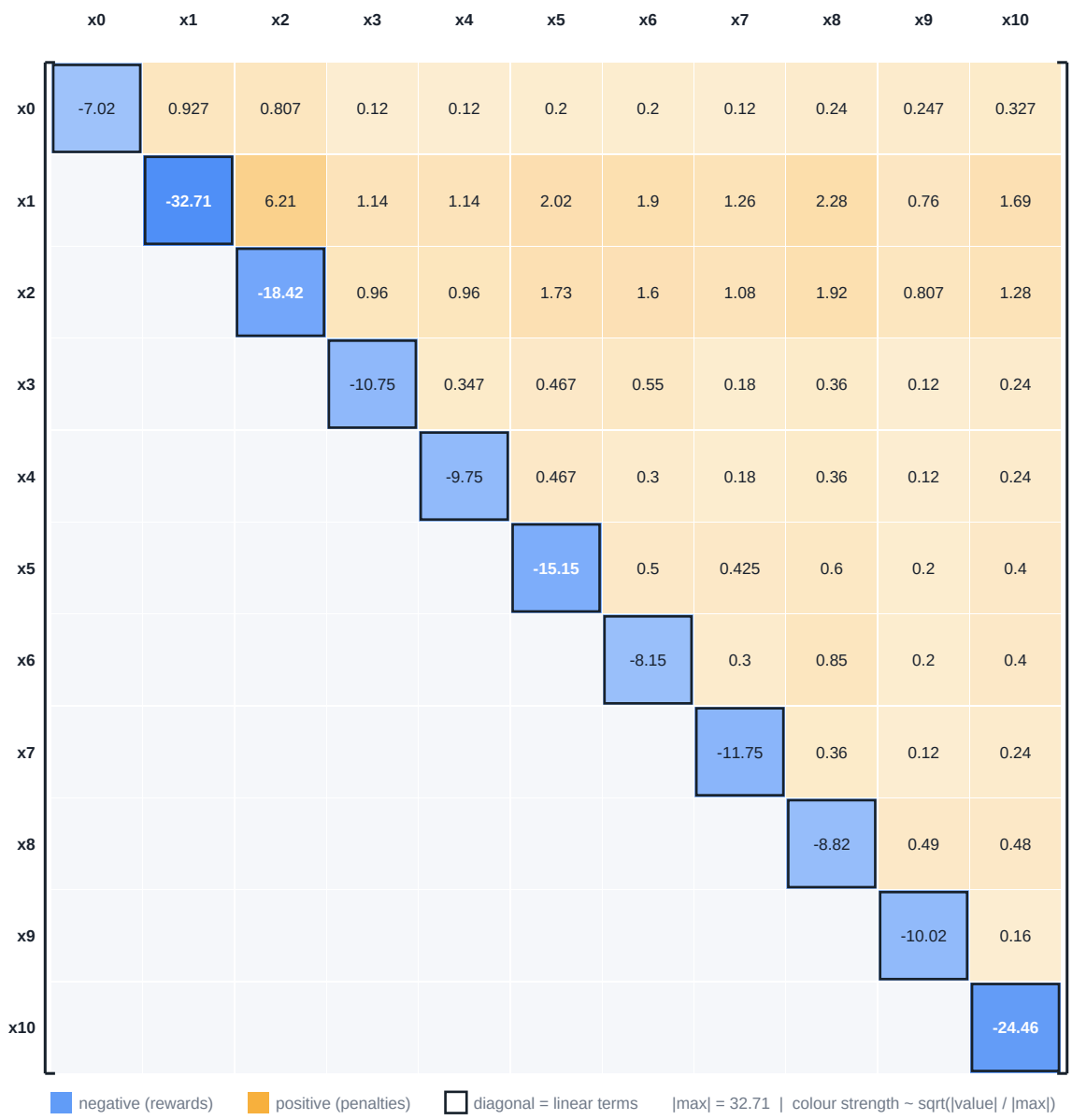
Formula

```
E = -(5 x0 + 21 x1 + 8 x2 + 8 x3 + 7 x4 + 11 x5 + 4 x6 + 9 x7 + 4 x8 + 8 x9 + 21 x10) - 1*(89 - (2 x0 + 19 x1 + 16 x2 + 3 x3 + 3 x4 + 5 x5 + 5 x6 + 3 x7 + 6 x8 + 2 x9 + 4 x10)) + 0.01*(89 - (2 x0 + 19 x1 + 16 x2 + 3 x3 + 3 x4 + 5 x5 + 5 x6 + 3 x7 + 6 x8 + 2 x9 + 4 x10))^2 + 0.25((1 - x0 - x1 - x10 + 0.666667*(x0*x1 + x0*x10 + x1*x10)) + (1 - x1 - x2 - x5 - x7 + 0.5*(x1*x2 + x1*x5 + x1*x7 + x2*x5 + x2*x7 + x5*x7)) + (1 - x0 - x2 - x9 + 0.666667*(x0*x2 + x0*x9 + x2*x9)) + (1 - x3 - x4 - x5 + 0.666667*(x3*x4 + x3*x5 + x4*x5)) + (1 - x10) + (1 - x3 - x6 + 1*(x3*x6)) + (1 - x4) + (1 - x8 - x9 + 1*(x8*x9)) + (1 - x7) + (1 - x6 - x8 + 1*(x6*x8)))
```

Term	Coefficient	Term	Coefficient	Term	Coefficient
x0	-7.02	x1	-32.71	x2	-18.42
x3	-10.75	x4	-9.75	x5	-15.15
x6	-8.15	x7	-11.75	x8	-8.82
x9	-10.02	x10	-24.46	x0*x1	0.9267
x0*x2	0.8067	x0*x3	0.12	x0*x4	0.12
x0*x5	0.2	x0*x6	0.2	x0*x7	0.12
x0*x8	0.24	x0*x9	0.2467	x0*x10	0.3267
x1*x2	6.205	x1*x3	1.14	x1*x4	1.14
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x1*x8	2.28	x1*x9	0.76	x1*x10	1.6867
x2*x3	0.96	x2*x4	0.96	x2*x5	1.725
x2*x6	1.6	x2*x7	1.085	x2*x8	1.92
x2*x9	0.8067	x2*x10	1.28	x3*x4	0.3467
x3*x5	0.4667	x3*x6	0.55	x3*x7	0.18
x3*x8	0.36	x3*x9	0.12	x3*x10	0.24
x4*x5	0.4667	x4*x6	0.3	x4*x7	0.18
x4*x8	0.36	x4*x9	0.12	x4*x10	0.24
x5*x6	0.5	x5*x7	0.425	x5*x8	0.6
x5*x9	0.2	x5*x10	0.4	x6*x7	0.3
x6*x8	0.85	x6*x9	0.2	x6*x10	0.4
x7*x8	0.36	x7*x9	0.12	x7*x10	0.24
x8*x9	0.49	x8*x10	0.48	x9*x10	0.16

QUBO matrix (11 x 11)

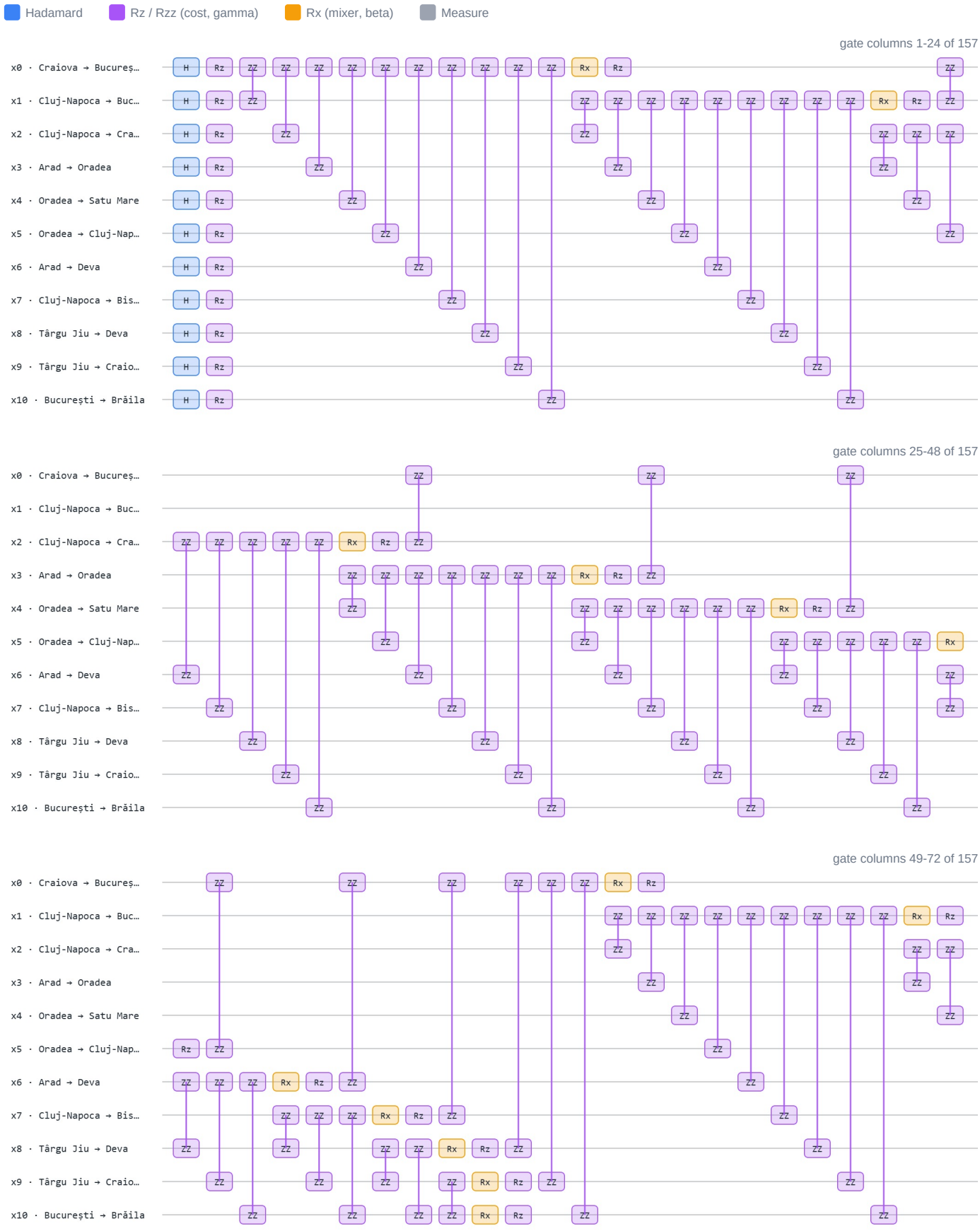
upper-triangular Q: row/column order = matrix_variables, diagonal Q[i][i] = linear coefficient of y_i, Q[i][j] (i < j) = coefficient of y_i*y_j, below the diagonal 0; E(y) = offset + y^T Q y (bits, so y_i^2 = y_i)



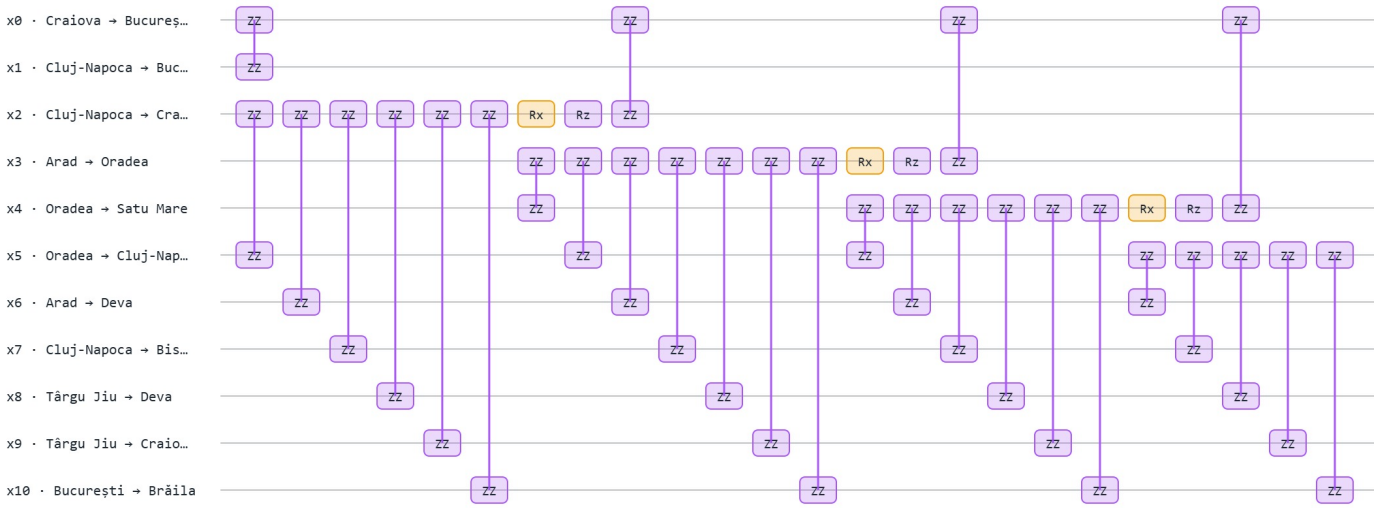
QAOA circuit

11 qubits | depth 49 | 165 two-qubit gates | rzz 165, rz 33, rx 33, h 11, measure 11

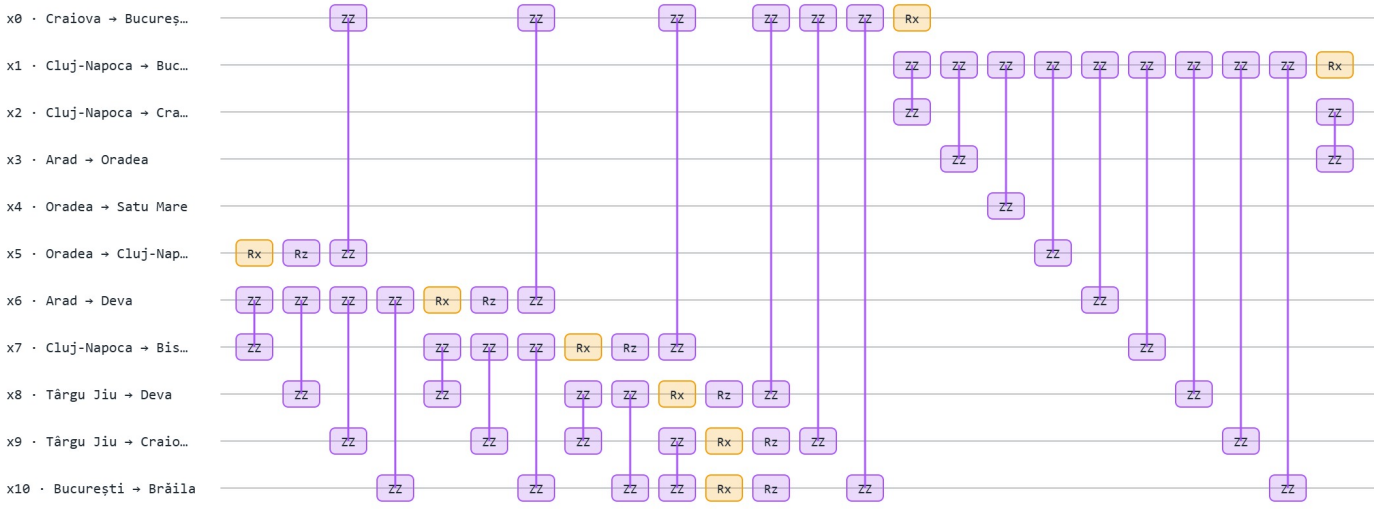
gamma = [-0.6870, -1.4704, -2.1087] beta = [0.8459, 0.3756, 0.2695]


Hadamard
Rz / Rzz (cost, gamma)
Rx (mixer, beta)
Measure
gate columns 1-24 of 157
gate columns 25-48 of 157
gate columns 49-72 of 157
x0 · Craiova → Bucureș…
x1 · Cluj-Napoca → Buc…
x2 · Cluj-Napoca → Cra…
x3 · Arad → Oradea
x4 · Oradea → Satu Mare
x5 · Oradea → Cluj-Nap…
x6 · Arad → Deva
x7 · Cluj-Napoca → Bis…
x8 · Târgu Jiu → Deva
x9 · Târgu Jiu → Craio…
x10 · București → Brăila

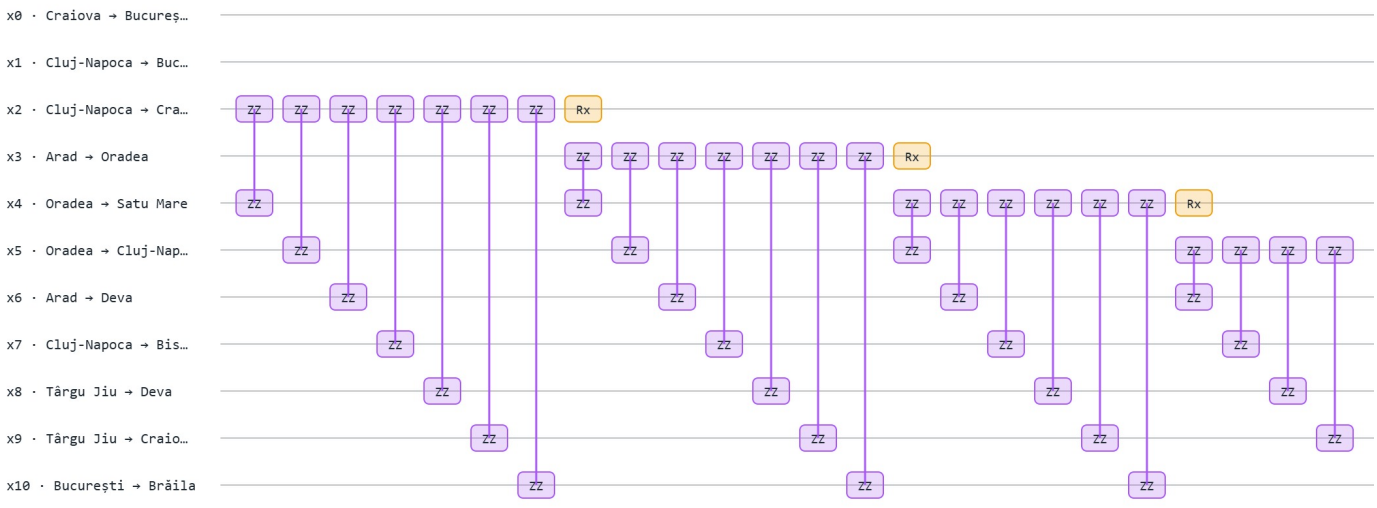
gate columns 73-96 of 157

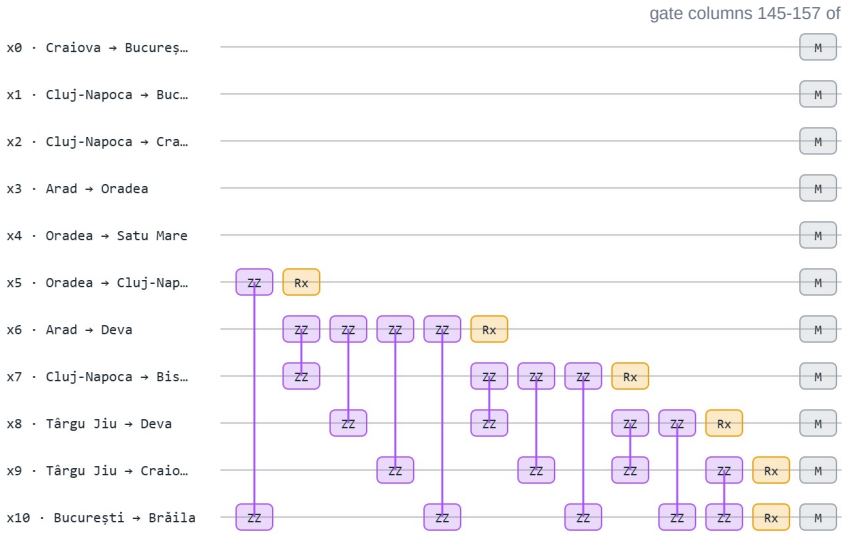


gate columns 97-120 of 157



gate columns 121-144 of 157




gate columns 145-157 of 157
x0 · Craiova → Bucureș…
x1 · Cluj-Napoca → Buc…
x2 · Cluj-Napoca → Cra…
x3 · Arad → Oradea
x4 · Oradea → Satu Mare
x5 · Oradea → Cluj-Nap…
x6 · Arad → Deva
x7 · Cluj-Napoca → Bis…
x8 · Târgu Jiu → Deva
x9 · Târgu Jiu → Craio…
x10 · București → Brăila
ZZ
Rx
M